QM47400 · Spring 2027 · Milestone 01

Initial Project Proposal

DUE	**Sunday, February 7, 2027, 11:59 PM ET**
OPENS	Wednesday, January 27, 2027
SUBMIT	One PDF per group, uploaded by one member: `NN_proposal.pdf` (NN is your group number, for example `03_proposal.pdf`)
GRADING	50 points: five criteria, 10 points each

What this milestone is for

This is where your group says what it will predict. Instead of a classic research question, you define a **prediction goal**: the specific outcome you will forecast from your data, and why getting it right matters to someone. Everything that follows (the outline, the abstract, the models, the poster) is built on these one or two pages.

What to do

Write a proposal of one to two pages with five parts, each under its own heading.

1. **Prediction goal.** One clear sentence naming the outcome you will predict. For example: *"Predict the likelihood of a customer churning within the next six months based on their transaction history and engagement metrics."*
2. **Motivation and significance.** Why this goal matters: the business problem or industry trend behind it, and what accurate predictions would change.
3. **Data overview.** Your project must use **real data**: observations of actual people, organizations, transactions, markets, public records, surveys, experiments, devices, or other real processes. Synthetic, simulated, artificially generated and AI-generated datasets are not eligible, and neither is the synthetic Bank Churn data of the Course Case Competition. Give all of the following:
   - a direct link to the dataset and the name of its original provider or collector;
   - what one row represents, and when and how the data were collected;
   - the response you will predict and the predictors you expect to use;
   - why the data suit a predictive project;
   - a clear statement that the dataset contains real observations and is not synthetic, simulated or generated.

   If the documentation does not say where the observations came from, choose another dataset or ask Professor Moreira before your group commits to it.
4. **Preliminary methods.** The techniques you plan to try, for example logistic regression, decision trees or neural networks.
5. **Expected contributions.** What you expect to achieve, and how your predictions could help a business decision.

Use clear headings and short bullet points, in clear, accessible academic language, and proofread before you submit.

What to submit

One member uploads `NN_proposal.pdf` for the whole group: one to two pages covering the five parts.

How it is graded

Fifty points: each of the five parts above is one criterion worth 10 points, scored as exemplary (9–10), proficient (7–8), developing (5–6) or beginning (0–4).

- **Prediction goal:** specific, measurable, and tied to a business use.
- **Motivation and significance:** a convincing reason, linked to a real business challenge or industry trend.
- **Data overview:** an eligible real dataset with its link, original provider, what a row is, how and when it was collected, the response and predictors, and why it fits the goal. A synthetic or generated dataset, or an overview that cannot be verified, falls in the beginning band.
- **Preliminary methods:** techniques clearly described and matched to the goal.
- **Expected contributions and organization:** clear value for decision-making, in a well-organized, well-formatted, error-free document.

Tips

- Use the vocabulary of notebooks 01 to 05: prediction goal, exploratory data analysis, leakage-safe splits.
- Check where the data come from before anything else. A dataset you cannot trace back to its source will cost you later.
- Keep this file: it seeds Milestone 02 (the expanded outline) and Milestone 03 (the draft abstract).

The Final Project Overview PDF has every date, the meeting rounds, and how the project is graded.